Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	Foot Reaction Forces During Simulated ISS Exercise Countermeasures		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	02/01/2004	**End Date:**	01/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Perusek, Gail (NASA GRC)		
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Magnitude of loads (F) and rate of change of loads (dF/dt) are widely believed to be important criteria for efficacy of countermeasures against loss of bone mass. However, it is known that astronauts on the first 6 expeditions to the ISS lost lower-extremity bone mass despite exercising. As the current exercise countermeasure devices (TVIS, CEVIS, iRED) are used and upgraded during their anticipated 10-year life span, there is a need to establish their loading characteristics in a realistic ground-based simulation so that exercise prescriptions and the design of components such as the harness and subject load device (SLD) can be optimized. In particular, the compliant nature of the interface between the device and its attachment to the ISS has an important influence on F and dF/dt and has not yet been adequately included in previous simulations.
The original aims of this project are: (1) To improve the biomechanical validity of the human supine suspension approach to microgravity simulation and to develop hardware and software that will characterize the force magnitude

Task Description:	(F), rate of change of force (dF/dt), and vibratory and transmission characteristics of load transferred to the human body. (2) To fully characterize lower-extremity loading during high-fidelity simulated 0-g exercise on treadmill, cycle ergometer, and resistance exercise devices throughout the range of SLD and harness settings. And (3) Based on this new information regarding mechanical “dose”, to propose a regimen of exercise on the available devices that will be effective in maintaining bone and muscle mass. To achieve the specific aims of the project, we will conduct modeling, analysis, and human subject experiments to determine the optimal biomechanical simulation of exercise in microgravity. The optimal simulation will take into account the mechanical and inertial conditions of the installation of exercise devices on the ISS. The findings from this specific aim will be generalizable to future countermeasure devices and future space vehicles. Based on these findings, we propose to enhance the existing Zero Gravity Locomotion Simulator (ZLS) at the Cleveland Clinic to provide a more flight-like simulation of exercise. The new simulator will then be used for a series of human subject experiments to define the entire range of musculoskeletal loading possibilities in cycling, treadmill running, and resistance exercise using existing and improved SLDs. Through a comparison of data collected from the simulator and previously measured daily loads in 1-g, we will provide clear guidelines for the prescription of in-flight exercise on existing devices that will protect against loss of bone and muscle mass based on a directly measured mechanical “dose.” In the first year of this project the team at CCF, NASA GRC and ZIN Technologies completed hardware design and finite element (FE) modeling for all major system components, and completed fabrication of 2/3 of these components for development of an Enhanced Zero-g Locomotion Simulator (eZLS). The team procured hardware for a servo-motor Subject Load Device (SLD) and completed the design of load splitters which will be used to distribute the gravity replacement loads to the subject’s shoulders and waist in varying ratios. The GRxF, against which the treadmill, cycle ergometer, or resistance exercise device may react, was analyzed by FEM to inform the design and attachment method to ground. FEM was also performed on the exercise device mounting framework / air bearing table assembly to ensure design was adequate to meet minimum non-rigid body modes. The mechanical design and FEM effort was paralleled with a dynamic modeling effort in MATLAB Simulink, to characterize system dynamic behavior. In Year 2 of the project, construction and system integration activities began at GRC and will be complete early in Year 3. The project team has begun development of a new feedback and control technology to achieve the constant load goal with the SLD. Other subject loading options are also being pursued including the development of a pneumatic SLD (pSLD). In addition to work on the SLD, a meeting was held with leaders of the backpack industry and consultation with JSC resulted in a new harness design. In the current year (Year 2), the dynamic mathematical model has progressed to include the human test subject and ISS vehicle dynamics. In the upcoming year (Year 3), installation and optimization of the eZLS (Specific Aim 1) will be complete. An appropriate SLD will be implemented with sufficient control technology and human subject testing will be performed to address Specific Aim 2. In this phase of the experiment we will systematically examine the potential mechanical input that the three main types of currently proposed exercise countermeasures can elicit. Due to hardware constraints, the project has fallen slightly behind schedule. However, modeling and analysis have been performed for Specific Aim 1 to inform the design of the enhanced Zero Gravity Locomotion Simulator (eZLS). Human subject experiments to optimize the biomechanical simulation of exercise in microgravity have yet to be performed. We anticipate completion of Specific Aim 1 in early 2006 (Year 3). Once optimized, experiments with all three exercise modalities to address Specific Aim 2 will be performed, with an anticipated start of March 2006 and completion prior to the end of Year 3. To address Specific Aim 3, engineers are modifying existing mathematical theories on Daily Load Stimulus to develop a comprehensive algorithm. This algorithm will be used in conjunction with the data collected under Specific Aim 2 in order to develop clear guidelines for exercise prescriptions. This Specific Aim will not be complete until Year 4.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge gained from this research will contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	Mechanical Design and Fabrication During the past year, the team at CCF, NASA GRC and ZIN Technologies completed hardware fabrication for development of an eZLS. Construction and system integration activities are complete at GRC. A complete system Stress Analysis Report was and positive safety margins were found for all components. Subject Load Device Servo-Motor System The project team continued development of the linear motor subject load device (SLD). The project team is developing and refining feedback and control technology to achieve the constant load goal, however, performance has not been optimal. Efforts continue but other options are being pursued in an effort to prevent the project from falling behind schedule. Pneumatic Subject Load Device In an effort to create a backup system for the Servo-Motor System, GRC and CCF collaborated to create a passive pneumatic subject load device (pSLD). This device uses a single pneumatic cylinder to provide a nearly constant spring force on the cables attached to the subject harness. Proof of concept testing was successfully completed at GRC and a duplicate system is on order for installation and validation at CCF. Delivery of the system is expected during mid-January 2006. Harness Design The SLD and harness are key components in the optimization of subject interface loading characteristics. Discomfort while wearing the harness directly impacts the efficacy of the exercise protocols performed. During this phase of the study, we explored design changes in the present harness to improve subject comfort. A meeting was held with leaders of the backpack industry and consultation with JSC resulted in a new harness design. Changes include, broadening of

	the shoulder straps and using an improved pelvic crest interface. Proof of concept testing has been completed at CCF. Additional funding from the John Glenn Biomedical Engineering Consortium (JGBEC) was awarded to support the fabrication of multiple harnesses for human subject experiments. System Dynamic Modeling A dynamic model of the treadmill rack and variable compliant interface hardware was integrated with a human foot impact model. Our integrated model results showed that with a vibration isolation system installed the peak reaction force on the foot could drop by up to 14% compared to a grounded system. The model will be enhanced with validated dynamic data and more realistic running conditions and then used to predict how prescriptions can be adjusted to reach the desired DLS in a reduced gravity environment. Exercise Prescriptions The project team is modifying existing mathematical theories on daily load stimulus (DLS) to develop a comprehensive algorithm. We formulated a revised DLS algorithm which accounts for the peak load as well as the rate of change of the bone load and other important factors. The revised algorithm is currently being implemented into a toolkit for the processing of test subject force-plate data.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Cavanagh, P R, Davis, B L, Rice, A J, Genc, K O, Maender, C C, Perusek, G P, Francisco, D R, Grodsinsky, C M, and Just, M L. "Lower-Extremity Loading During Simulated Microgravity and Long-Duration Space Flight." . 2005 May;. , May-2005
Abstracts for Journals and Proceedings	Just, M L, Grodsinsky, C, Perusek, G, Davis, B, and Cavanagh, P. "Modeling and Simulation of Reaction Forces in a Reduced Gravity Exercise System." . Published. , Jan-2006
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Articles in Peer-reviewed Journals	Cavanagh, P. R. "Exercise and Bone Health in Space" Jan-2006